

SOP 201: MRI Safety Screen

1 Summary

All individuals must be screened before they enter the Magnet Room. This document establishes MRI Safety screening procedures at the Notre Dame Human Neuroimaging Center (ND-HNC).

2 Scope

These policies apply to all individuals who enter Zones 3 & 4 (B15, B17, B17a; Fig. 1). Emergency response personnel are exempt during an emergency response **ONLY IF** the door to the Magnet Room is closed and locked with a key.

3 Definitions

- **Magnet Room:** B17A; The room within the MRI suite that contains the MRI scanner. It is accessed from within the MRI Control room (B17) through a doorway that is controlled by an electronic lock.
- **ND-HNC:** Notre Dame Human Neuroimaging Center, Lower Level of Veldman Family Psychology Clinic.
- **Operator:** MRI Technologist or Level 4 researcher, responsible for conducting research operations and maintaining equipment. Typically the Lead during emergency responses.
- **Visitor:** An individual who has not taken the ND-HNC MRI Safety Course but has a legitimate reason for entering the MRI suites. Research participants and their legal guardians are Visitors. Other Visitors may include: individuals providing/receiving a tour of the VFPC and ND-HNC, faculty and staff who have not received MRI Safety Course, and research assistants.

4 Notes

None.

5 Policies & Procedures

The purpose of MRI Safety screening is to determine whether entry into the Magnet Room poses a health or safety risk. Accordingly, all individuals entering Zones 3 & 4 must be screened according to the procedures below.

5.1 MRI Prescreen

Prior to the Visitor's arrival at the ND-HNC, research personnel conduct a prescreen to ensure that the Visitor does not have any MRI contraindications:

- Non-MRI compatible medical implants.
 - **Note:** Visitors with an implant must present documentation from their surgeon's office that the device is MRI compatible.
 - **IUDs:** North American non-metallic intrauterine devices (e.g. Mirena) are approved. Metallic (e.g. Cu) and IUDs from other regions must be approved by the MRI Technologist.
- Other metal embedded in the body (e.g. shrapnel, unremovable piercings).
- Pregnancy.
- Claustrophobia (the Mock can be used to assess comfort levels).
- Tattoo on face or neck (including permanent makeup).

5.2 MRI Preparation

To facilitate efficient operations and MRI compatibility, the Visitor should be informed of the following restrictions to guide their preparations to enter the Magnet Room:

- Clothing should be free of metallic objects and clasps; bras should not have an underwire. Anti-odorant materials (e.g. Lululemon's Silverescent) are not MRI compatible.
- Jewelry will need to be removed, including piercings if they are metallic.
- Cosmetics such as colored contacts, magnetic lashes, and makeup or nail polish with glitter will have to be removed.
- Hair must be free of metallic clips, barrettes, etc.

5.3 MRI Safety Screen

All who enter Zone 3, regardless of whether they will enter Zone 4 (Magnet Room) fill out Form 001: MRI Safety. ND-HNC staff and research personnel submit a form (a) annually

and (b) whenever a change occurs in their MRI compatibility. Visitors fill out the form in Zone 1, and researchers with at least Level 2 access should be available to answer any questions.

- For Visitors, Form 001 must be filled out on the same day as entering Zone 3.
- Each question/prompt must be answered individually.
- In the case of minors, their legal guardian must fill out the form.
 - Note: verify guardianship, which may differ from the individual who brought the minor to the center.
- The Visitor signs and dates the form in the designated location.

5.4 MRI Safety Screen, Review 1

A researcher with at least Level 2 access will review the Visitor's responses on Form 001.

- Discuss any potential contraindications, points of confusion, or concerns. Inform the Operator about any items discussed or addressed.
- After reviewing Form 001, the researcher prints and signs their name in the designated location.
- If the researcher determines the Visitor is MRI compatible, they may escort the Visitor into Zones 2 & 3.

5.5 MRI Safety Screen, Review 2

A second review is required for Visitors entering Zone 4 (Magnet Room).

- The Operator conducts the second review, who must be a different from the person who conducted Review 1.
- The Operator signs and dates Form 001.
- The Visitor may enter Zone 4 only after the Operator has determined that it is safe to do so.
- ND-HNC only stores MRI Safety forms for staff and researchers. Visitor forms are maintained by their respective research teams.

6 Figures



Figure 1: Schematic of ND-HNC MRI Zones. **Orange** Zone 1. **Purple** Zone 2. **Green** Zone 3. **Blue** Zone 4.